

Employee Attrition Prediction – Executive Summary

Objective

The objective of this project was to identify employees who are more likely to leave the organization by analyzing workplace and employee-related factors. Early identification of at-risk employees enables HR teams to implement effective retention strategies.

Key Findings

The overall employee attrition rate is approximately 16%, indicating that most employees remain with the organization while a smaller proportion leave.

Employees who work overtime, earn lower monthly incomes, and have fewer years at the company are more likely to leave.

Certain departments and job roles show higher attrition rates and should be prioritized for retention initiatives.

Salary alone does not explain employee attrition. Factors such as work-life balance, job satisfaction, and overtime also play significant roles.

Business Recommendations

Introduce workload management and flexible work arrangements to reduce excessive overtime and employee burnout.

Focus retention efforts on departments and job roles with higher attrition by providing career development opportunities, mentoring, and regular employee engagement programs.

Model Limitation

The model is based only on historical employee data and cannot capture personal reasons, organizational culture, or external job market conditions. Therefore, it should be used as a decision-support tool rather than the sole basis for HR decisions.

Conclusion

The project demonstrates that machine learning can help organizations identify employees at risk of leaving and support HR teams in making informed, proactive decisions to improve employee retention and workforce planning.